

Test Report – Shapes Calculator

1. Objective

Identify issues with:

- Validation of input
- Calculation accuracy
- Usability & system behaviour

2. Test Summary

System is stable and responds gracefully to errors. System redirects user with meaningful error messages when invalid input is given or invalid operations are attempted. System does not crash, and user is able to navigate menus appropriately, successfully adding, listing, and performing calculations on shapes.

However, a bug relating to invalid input and the Triangle class was identified when an area calculation gave a 0 result.

3. Test Results

The Triangle class caused an incorrect output when an invalid triangle was added and an area calculation was attempted.

All other objectives proved successful.

4. Defects

One defect with the Triangle class was identified. This class did not account for the unique requirement of the shape to be mathematically possible. Thus, impossible values (such as 1, 2, 3) could be input, causing the later area calculation to fail.

5. Conclusion

The system is largely working as expected and all test cases passed successfully. Functionality has been added within the Triangle class to further validate the mathematical possibility of the values given, and the test suite has been updated appropriately.